

Proof. If $n \in \mathbb{N}$ and $3n^2 + 5$ is odd. Then n must be even.

For the sake of contradiction, $n \in \mathbb{N}$ is odd. Suppose: $3n^2 + 5$ is odd.

If n is odd. Then we can write it in the form $n = 2k + 1, k \in \mathbb{N}$. This follows:

$$\begin{aligned} 3n^2 + 5 &= 3(2k + 1)^2 + 5 \\ &= 3(4k^2 + 4k + 1) + 5 \\ &= (12k^2 + 12k + 3) + 5 \\ &= 12k^2 + 12k + 8 \end{aligned}$$

Observe we can simplify this to the following term:

$$\begin{aligned} &= 12k^2 + 12k + 8 \\ &= 4(3k^2 + 3k + 2) \\ &= 2(2(3k^2 + 3k + 2)) \end{aligned}$$

Let: $(2(3k^2 + 3k + 2)) = m$

Note: All values in all prior operations are in $\mathbb{N}$ and all operations preserve closure $m \in \mathbb{N}$

$$2(2(3k^2 + 3k + 2)) = 2m = 3n^2 + 5$$

Thus $3n^2 + 5$ is even which is a contradiction thus the original statment "if $n \in \mathbb{N}$ and $3n^2 + 5$ is odd. Then n must be even." must be true. $\square$